

Perform and write study of Cryptocurrency development.

Study of Cryptocurrency Development

Aim

To study the development and working of cryptocurrency using blockchain technology.

Theory

What is Cryptocurrency?

Cryptocurrency is a digital currency that uses cryptography and blockchain technology for secure transactions.

It is decentralized, meaning no bank or central authority controls it.

Example:

- Bitcoin
 - Ethereum
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Features of Cryptocurrency

1. Decentralized system
 2. Secure transactions
 3. Fast money transfer
 4. Transparent records
 5. Digital ownership
 6. Blockchain based
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Steps in Cryptocurrency Development

Step 1 — Create Blockchain Network

A blockchain network is created to store transaction records.

Step 2 — Create Digital Coin

Digital currency or token is developed using blockchain.

Example:

- Bitcoin
 - Ethereum
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Step 3 — Use Cryptography

Cryptography secures transactions and wallets.

Step 4 — Wallet Creation

Users create crypto wallets to store coins.

Example:

- MetaMask
 - Trust Wallet
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Step 5 — Mining or Validation

Transactions are verified using mining or validators.

Step 6 — Smart Contracts

Smart contracts automate cryptocurrency operations.

Step 7 — Transaction Processing

Users send and receive cryptocurrency through blockchain network.

Applications of Cryptocurrency

1. Online payments
2. International money transfer
3. Smart contracts
4. Digital assets
5. Decentralized applications

Advantages

1. High security
2. No middleman
3. Fast transactions
4. Transparency
5. Global access

Disadvantages

1. Price volatility
2. Illegal activity risk
3. Requires internet
4. Difficult recovery if wallet lost

Simple Architecture

User → Wallet → Blockchain Network → Transaction Verification → Block Added

Output

Studied cryptocurrency development process and blockchain-based transaction system.

Conclusion

The development and working of cryptocurrency using blockchain technology was studied successfully.

Perform and write study Chain code in Hyperledger Fabric

Study of Chaincode in Hyperledger Fabric

Aim

To study Chaincode in Hyperledger Fabric blockchain platform.

Theory

What is Hyperledger Fabric?

Hyperledger Fabric is an open-source blockchain platform used for developing business blockchain applications.

It is permissioned blockchain, meaning only authorized users can access the network.

What is Chaincode?

Chaincode is a smart contract in Hyperledger Fabric.

It contains business logic that controls transactions and data in blockchain network.

Chaincode is written in:

- Go
 - Java
 - JavaScript
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Features of Chaincode

1. Executes business logic
 2. Manages ledger data
 3. Secure transactions
 4. Runs on blockchain peers
 5. Supports smart contracts
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Working of Chaincode

Step 1 — User sends request

User requests transaction through application.

Step 2 — Chaincode executes

Chaincode processes the request.

Step 3 — Data validation

Peers validate transaction.

Step 4 — Ledger update

Blockchain ledger gets updated.

Simple Chaincode Example

```
class SimpleContract {  
  
  async createAsset(ctx, id, value) {  
  
    const asset = {  
      ID: id,  
      Value: value,  
    };  
  
    await ctx.stub.putState(id, Buffer.from(JSON.stringify(asset)));  
  }  
  
}
```

Explanation

putState()

stores data on blockchain ledger.

Components of Hyperledger Fabric

1. Peer
 2. Orderer
 3. Ledger
 4. Channel
 5. Chaincode
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Output

Studied chaincode and its working in Hyperledger Fabric blockchain network.